UHAS

Universal Home Automation System

for

FIBARO Home Center 2

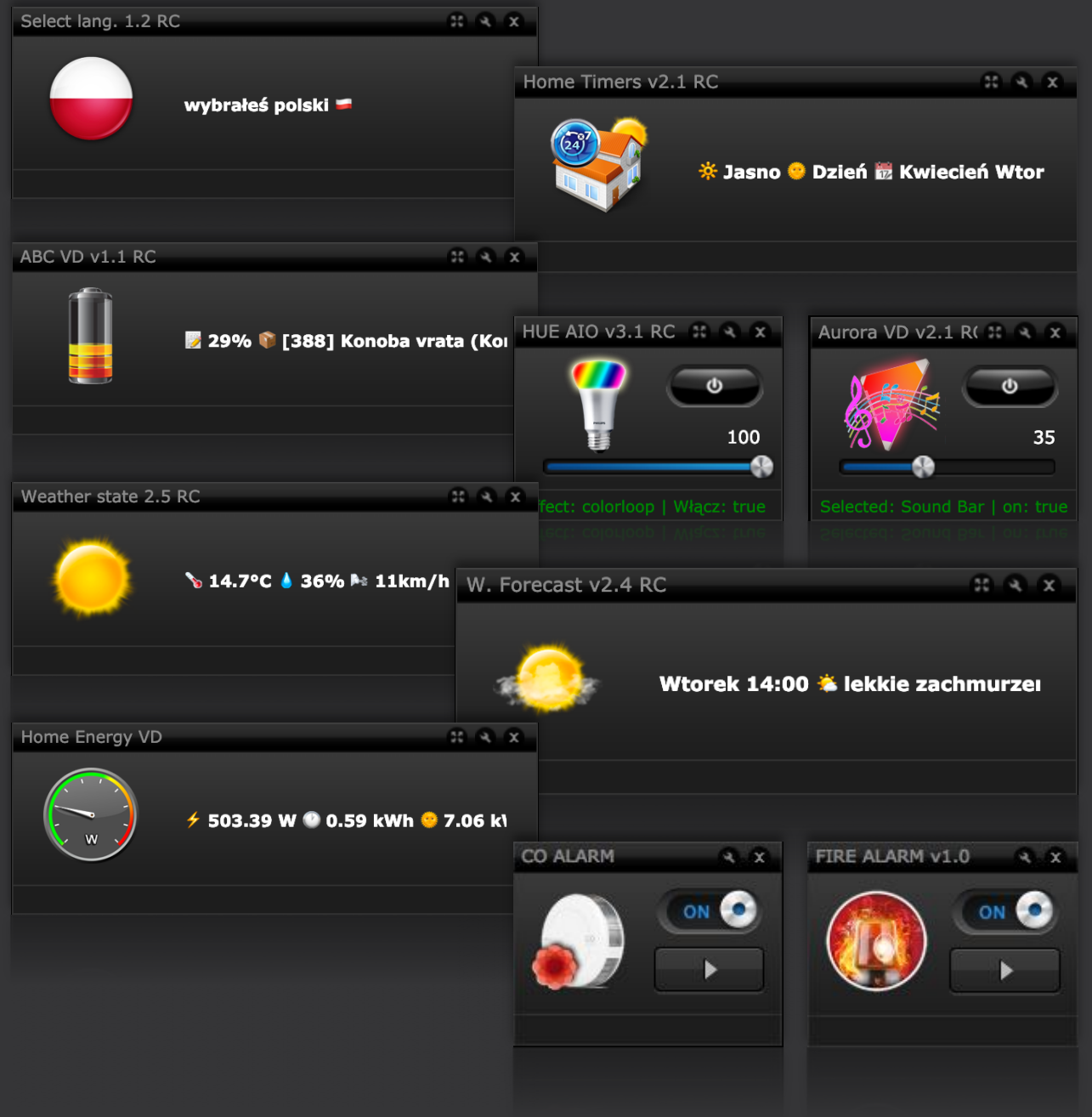


Presentation 3nd edition

What is UHAS

Set of scenes and Virtual Devices that take care of the following:

- Auto translation of all scenes, VDs, predefined global variables and messages to 27 different languages (all supported by HC2 + 6 more)
- Scheduled and repeating events based on time of day, week, month and season
- Lighting and blinds automation
- Weather state and forecast with notifications
- Presence monitoring
- Alarm Clock and Reminders
- Home energy consumption
- Appliance monitoring
- Alarms control (burglar, flood, fire and CO)
- Irrigation control based on FIBARO devices
- HC2 system status notifications and many more…



UHAS prerequisites

- FIBARO Home Center 2 with FW version 4.520 and above
- Several z-wave devices included



UHAS supported devices







- Netatmo weather station with rain and wind sensors
- Netatmo Welcome camera with presence monitoring and personalized notifications
- Philips HUE bridge with all types of bulbs
- Sonos speakers
- Nanoleaf Aurora & Canvas lights
- Smarter Coffee machine







Why UHAS ?

- UHAS offers basic part of the HA ready to be used out of the box with minimum user effort to setup. Running code is already there doing most of the settings for the user or for system installer.
- UHAS is developed in modules that are responsible only for one particular action or job and can interact with each other or exist on their own.
- UHAS is primarily intended to be used by new users of FIBARO system who want to have decent HA without need to learn LUA programming and spending a lot of time with setting it up.
- UHAS can also became part of matured FIBARO system installations which are grown with many modules that cover more advanced HA and more devices.

What are the UHAS modules?



UHAS Translation Module

UHAS translation module consists of two components:

- **Global Translation Scene** – keeps translations for predefined global variables and does translations of all scenes and VDs. During first run it generates global variable table with all translations.
- **Select Language VD** – checks HC2 default language and if it is changed initiates translation. It also gives users option to select additional 8 languages that are not supported by HC2 at the present time.



Supported languages HC2:

English	= "en"	Polski	= "pl"
Deutsch	= "de"	Svenska	= "sv"
Portugues	= "pt"	Italiano	= "it"
Francais	= "fr"	Nederlands	= "nl"
Roman	= "ro"	Brazilian Portuguese	= "br"
Estonian	= "et"	Latvian	= "lv"
Chinese	= "cn"	Russian	= "ru"
Denmark	= "dk"	Finland	= "fi"
Czech	= "cz"	US English	= "us"
Spanish	= "es"		

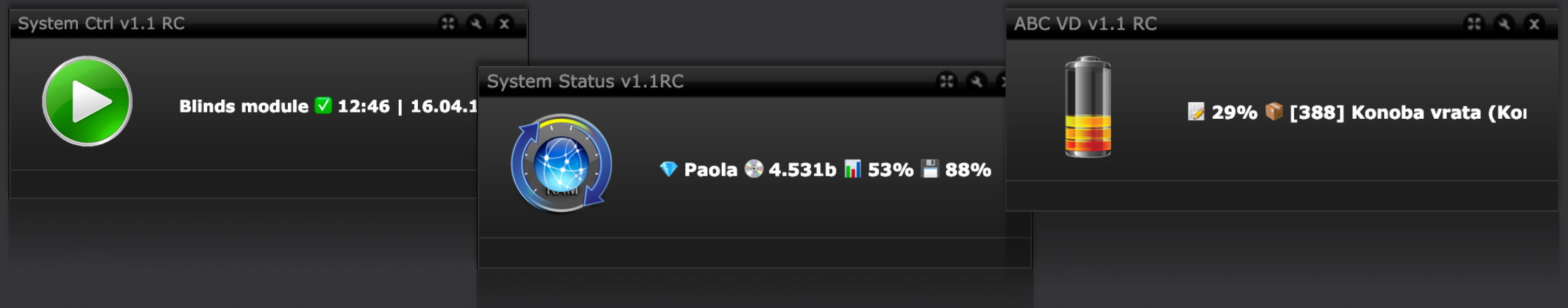
Additional languages supported by this version:

Slovak	= "sk"	Croatian	= "hr "
Bosnian	= "ba "	Serbian	= "rs"
Slovenian	= "si"	Norwegian	= "no"
Hungarian	= "hu"	Bulgarian	= "bg"

UHAS System Module

UHAS system module consists of several components:

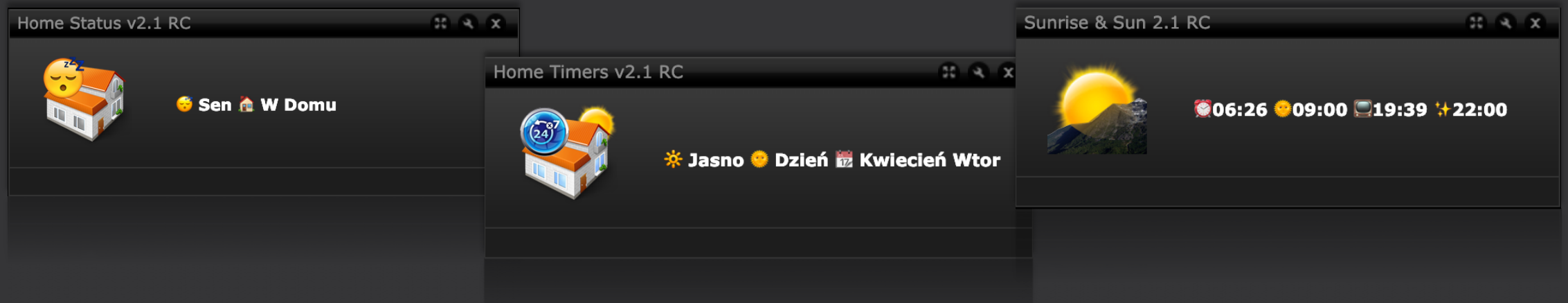
- **System Control VD** – allows user to stop/start running one UHAS module or complete system for upgrading firmware or changing devices setup. After restarting, UHAS will scan all changes and update
- **System Status VD** – allows user to quickly see current status of the system like free memory, free disk space, availability of internet connection and if there is available firmware upgrade.
- **Advanced Battery Check VD** – scans battery status of all battery operated devices and sends notification if any device will require battery replacement
- **Devices Status VD** – scans all devices and sends reports if any device is found “dead” or didn’t change status for defined time.



UHAS Scheduling Module

UHAS scheduling module consists of several components:

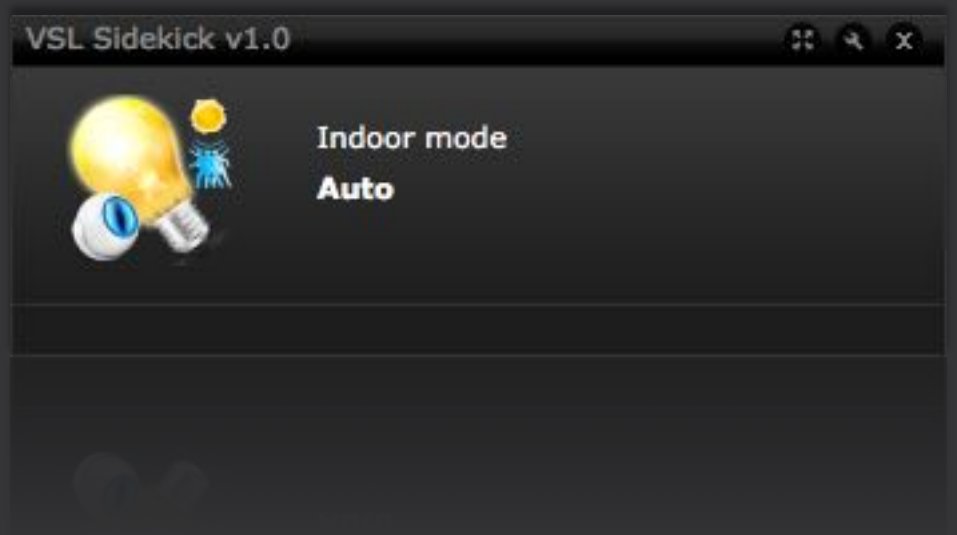
- **Main Scene FTBE** – (For Time Based Events) this scene takes care of all timers like Darkness, TimeOfDay, Weekday, Month, Season, Sunrise and Sunset Times etc. It can be also setup to execute other scenes, VDs, devices and change values of global variables at scheduled time or repeat at defined intervals.
- **Sunrise & Sunset VD** – allows user to setup at what time the time of day is changed to Morning, Day, Evening or Night. One can choose to follow sunrise and sunset for geographic location or at fixed time.
- **Home Status VD** – displays current status of sleep state and presence state. It also allows Demo Mode to be turned on for test or presentation purposes.
- **Home Timers VD** – displays current status of Darkness, Time of Day, Weekday, Month, Season and when seasons are changed. Similar as Home Status VD it allows to turn on Demo Mode.



UHAS Lighting Module

UHAS lighting module consists of two components:

- **VSL Scene** – (Very Smart Lights Scene) this scene should be installed one for each room. It can control all lights of that room triggered by motion sensor, door sensor and/or global variable. It supports dimmer modules, switch or relay modules, wall plugs, Philips HUE plugin and VD's that control Philips HUE. It has many ways how it can be setup. Details are to be covered in a separate manual "User Guide to VSL Scene"
- **VSL Setup VD** – is responsible to add all necessary global variables used by VSL Scene. Also allows user to setup lights and sensors. User can simply add or change automatic and manual lighting moods by first setting all lights to their brightness levels and color and then save that setting.

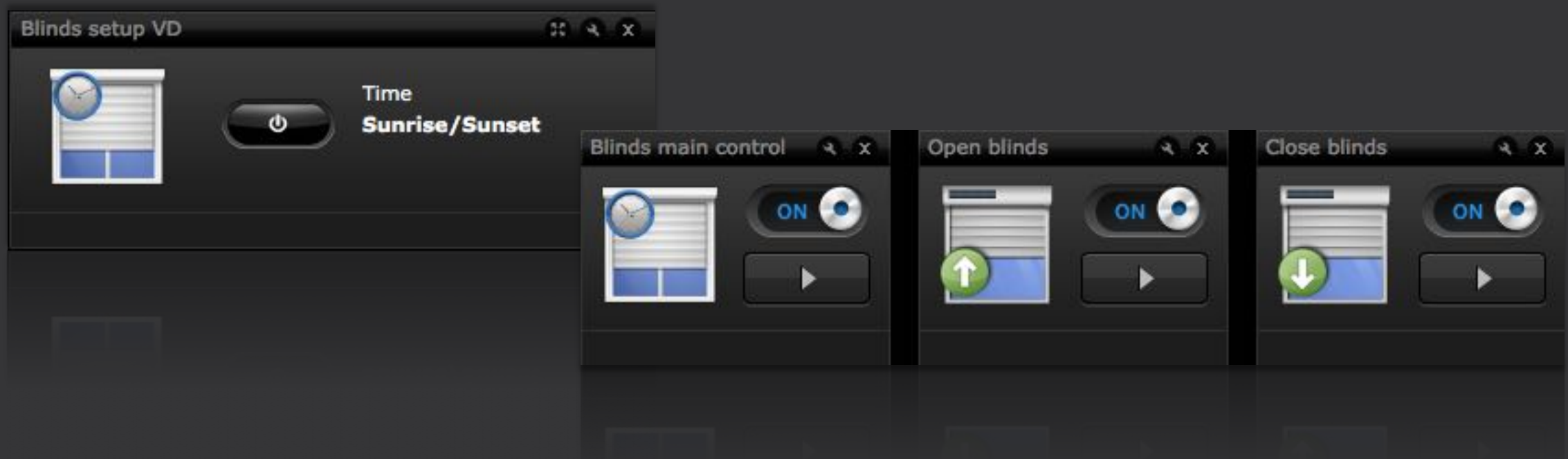




UHAS Blinds Module

UHAS blinds module consists of two components:

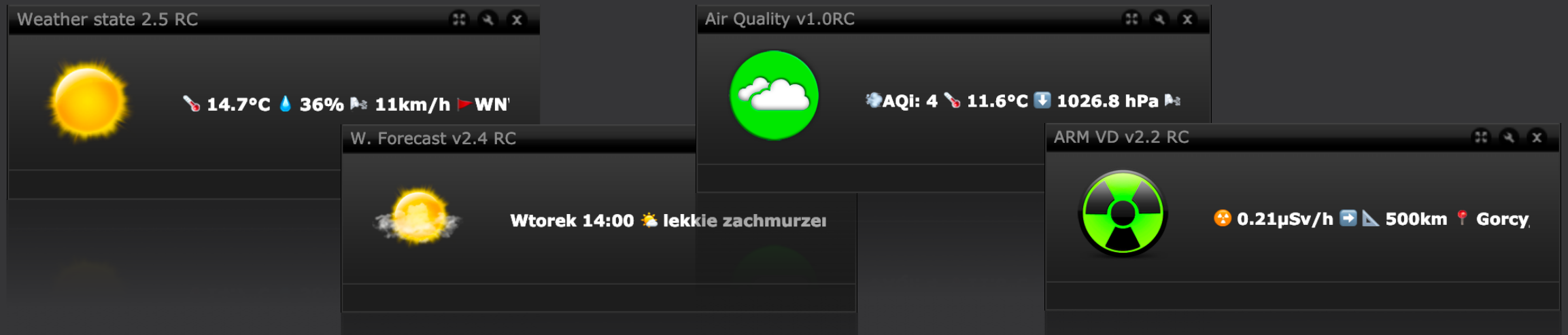
- **Blinds Control Scene Group** – this scenes control blinds for whole household depending on user settings made with Blinds Setup VD. It may also control depending on weather state (early development stage) .
- **Blinds Setup VD** – is used by user to define times when blinds open or close. E.g. at sunrise and sunset or wakeup and go to sleep defined times. Also it can close blinds for cinema room during day light.



UHAS Weather Module 1/2

UHAS weather module consists of minimum two components:

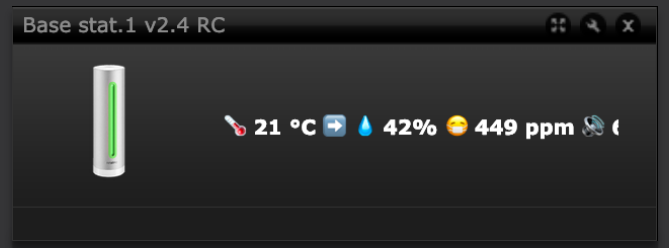
- **Weather State VD** – more than one Weather State VD can be setup to get weather status from more than one source! It supports two weather services: openweathermap.org and darksky.net. Main or first VD updates also weather on HC web GUI interface by means of weather provider plugin. The only user's setup is API key from weather provider. There are lots of icons to be setup. However it is now very easy with help of VD Icon Preview – which is also part of the UHAS extra module. Weather State VD also supports Netatmo weather station.
- **Weather Forecast VD** – same as above, more than one Weather Forecast VD can be setup to provide weather forecast from different sources. This VD may also send notifications. It supports same services as Weather State VD.
- **Weather Forecast VD** – same as above, more than one Weather Forecast VD can be setup to provide weather forecast from different sources. This VD may also send notifications. It supports same services as Weather State VD.
- Air Quality VD – user can now monitor quality of air in his neighborhood. Nowadays quality of air is very important and can influence our health so this VD will send notification if pollution levels rise.
- Advanced Radiation Monitor VD – monitors radiation levels for 5 nearest measuring stations and also shows station with the highest radiation level from all around the world. Will send notification alert if elevated radiation is detected on one of the nearest stations.

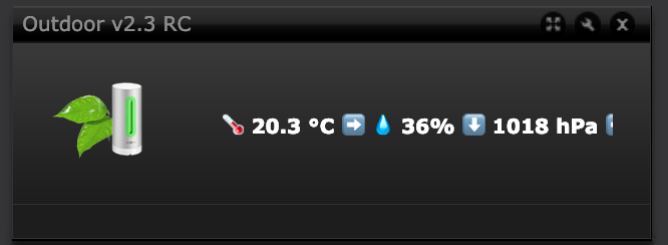


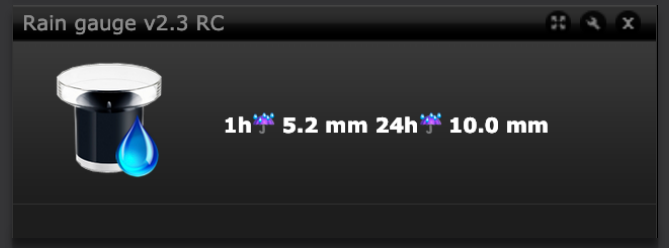
UHAS weather module 2/2

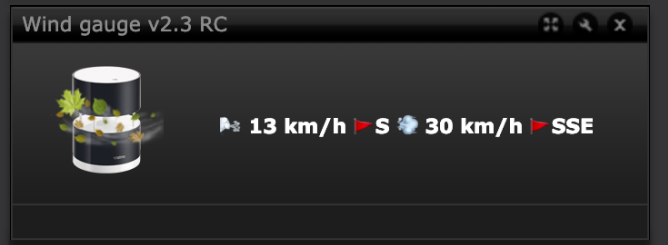
UHAS weather module fully supports Netatmo weather station with rain and wind modules included and consists of several components:

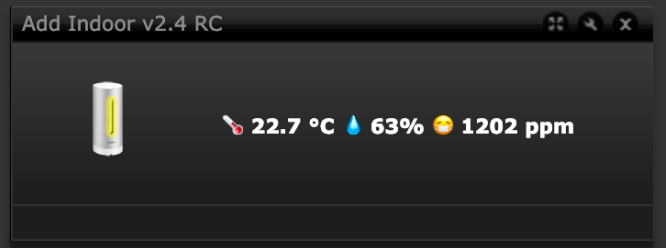
- **Netatmo Weather Station Scene** – supports more than one Netatmo weather station. The user needs to setup client ID, client secret, user name, password and station(s) ID. Global variable that store collected data is added to the system by Netatmo main indoor VD. This VD also takes care that weather is updated as soon as it is updated on Netatmo cloud.
- **Netatmo Module VD's** – Each Netatmo module has it's own VD to display data as Indoor Main Module VD, Outdoor Module VD, Rain Sensor VD, Wind Sensor VD and Additional Indoor Module VD. They are all updated by Netatmo main indoor VD to show latest readings.

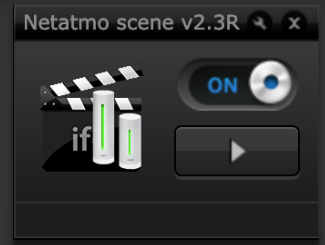








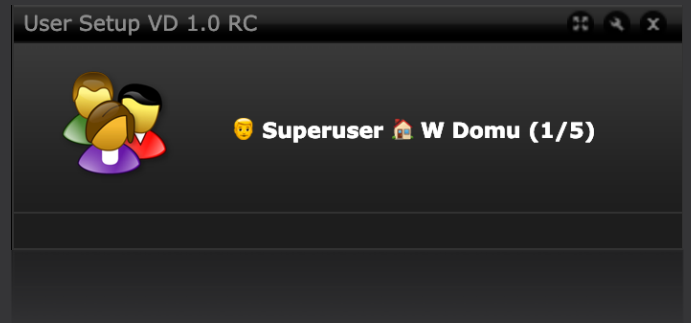


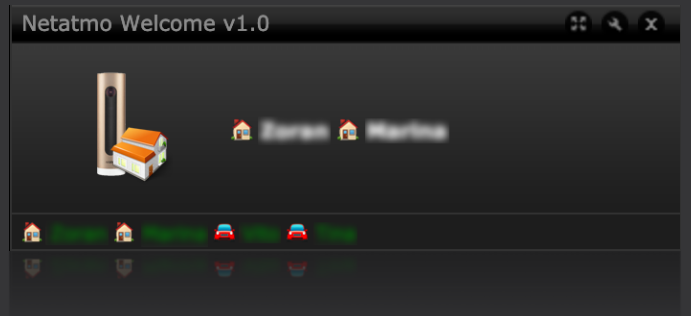


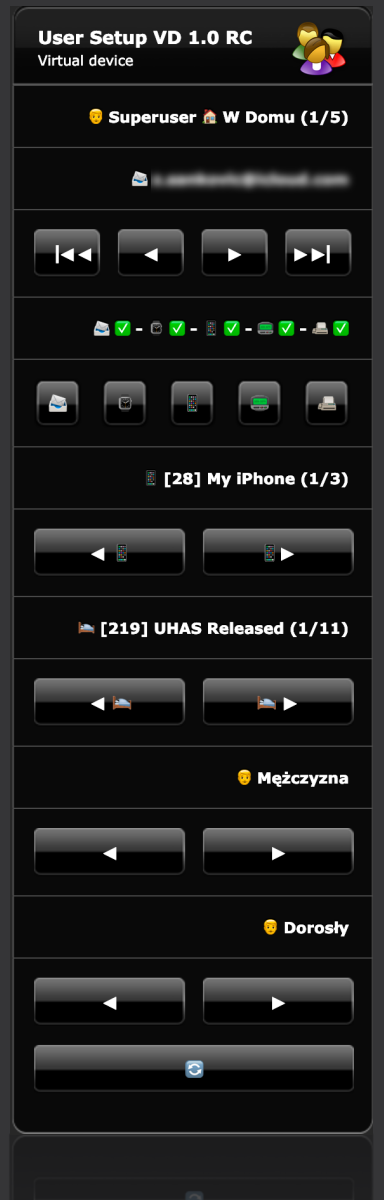
UHAS Presence Module

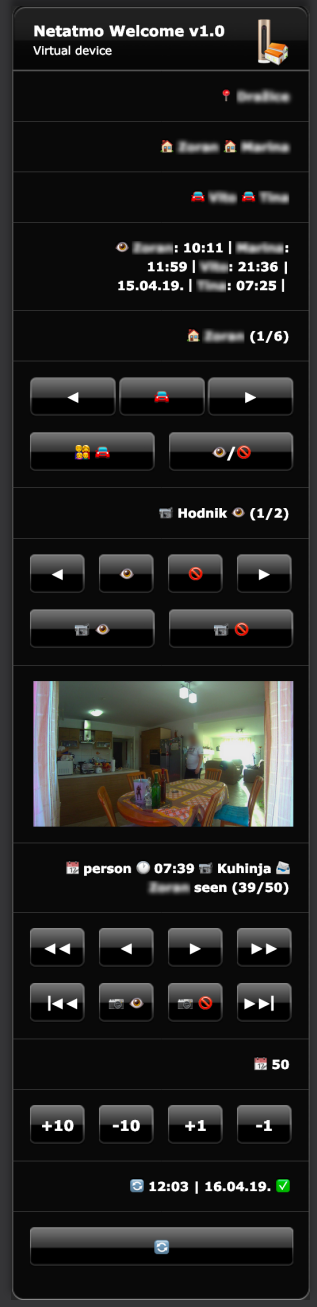
UHAS presence module monitors presence of household members:

- **HC User Setup VD** – Allows users to enhance default settings available on HC2 by adding gender and role to users and favorite room. Also can bond favorite mobile device and quickly change what type of notifications want or does not want to receive from system
- **Netatmo Welcome VD+scene** – Integrates Netatmo Welcome camera with HC2 allowing users to completely automate presence system with geofencing and facial recognition which allows automatic control of complete household and thanks to facial recognition allows HC2 to give more personalized responses.









UHAS Energy Module

UHAS energy consumption module is still in development but at the moment consists of two components:

- **Home Energy Scene** – scene main task is to update global variable with measurements and update VD.
- **Home Energy VD** – it is responsible to add all necessary global variables and to show measured consumption.
  NOTE: at the moment this is very simple setup. It is quite complex to add tariff system for so many different countries to properly calculate energy consumption in local currency.



UHAS Multimedia Module

UHAS multimedia module is still in development but at the moment consists of several components:

- **Remote control VD's** – with predefined buttons and code ready to work with iTach devices and support for HC2 plugins
- **Sonos Remote VD** – based on famous Sonos Remote VD developed by J.C. Vermandé now modified to be part of the UHAS. Supports TTS (text to speech)
- **Sonos API Player VD** – player that uses jishi node Sonos API interface. Supports several TTS services.



UHAS security and safety Modules

UHAS security and safety modules are still in development at the moment and will consists of several components:

- **Burglar Alarm Setup VD** – user use this VD to setup behavior of the alarm system and also select what sensors are armed for partial arming. This VD also is responsible for activating Mimic home presence scene that controls VSL scenes to turn on/off lights to pretend user is at home.
- **Burglar Alarm Arming Scene** – scene that is responsible to check door/window status and arm all selected sensors.
- **Burglar Alarm Disarming Scene** – scene that is responsible to disarm all selected sensors.
- **Flood Alarm Scene** – scene that is responsible to send notifications and do some other actions defined by user in the case that leakage is detected
- **Fire Alarm Scene** – scene that is responsible to send notifications and do some other actions defined by user in the case that fire is detected
- **CO_2 & CO Alarm Scene** – scene that is responsible to send notifications and do some other actions like sound alarm in the case that concentration of CO_2 and CO is above threshold for too long



UHAS Appliance Module

UHAS Appliance Module consists of two components:

- **Appliance Monitoring Scene** – Scene main task is to monitor appliance consumption and to send notification when appliance started/finished working. Need to install one scene per appliance.
- **Appliance Monitoring VD** – is responsible to add all necessary global variables and to show status of the appliance and measured energy consumption. It can also be used to set schedule start and to turn on/off appliance remotely, One VD per appliance is needed.

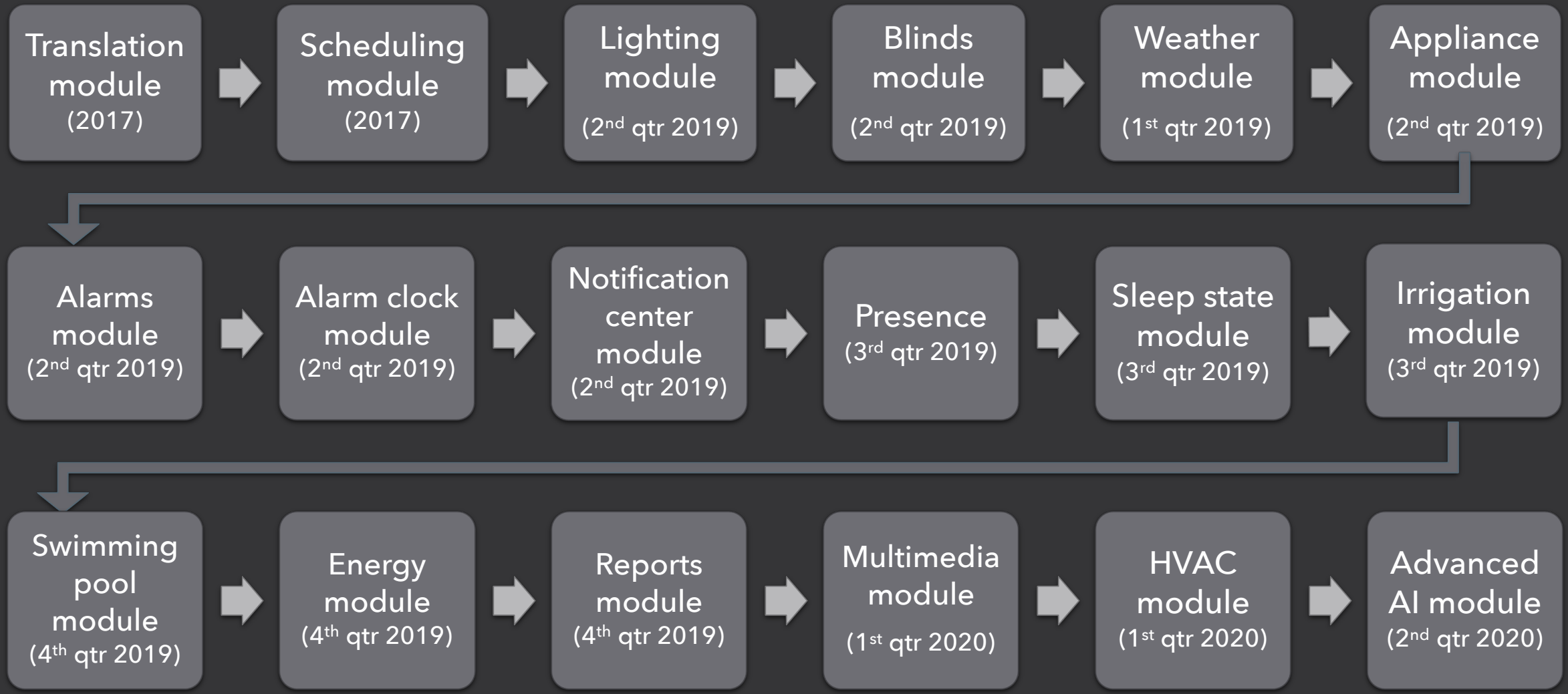


UHAS Other Modules

UHAS has other modules. Some of them are finalized and some are in early stage of development:

- **Notification module** – Centralized control and setup of all notifications will allow user to configure for each message what format to use (e-mail, HC2 push, HC2 popup, Pushover, Telegram and TTS or any combination of supported formats. Also will support queueing of notifications that could not be sent due to internet connection brake down and also VD to check queued and delete queue. Remainders will be included in this module (holidays, birthdays and other reminders).
- **Swimming Pool module** – Control of swimming pool pump, monitoring chemicals levels with notification when drop low, measuring temperature of water. Available control of pump manual, automatic (based on water temperature) and Timer with set of periods that user can setup.
- **Irrigation module** – control of irrigation system based on Fibaro relay modules or integration with other manufacturers systems like Rachio.
- **HVAC module** – Control of air conditioning devices and heating. Still in development.
- **Alarm clock module** – alarm clock completely configurable by user, control of wake up music, lights and notifications like todays reminders, weather forecast etc. over TTS system
- And some other modules that are in development stage.

UHAS Roadmap



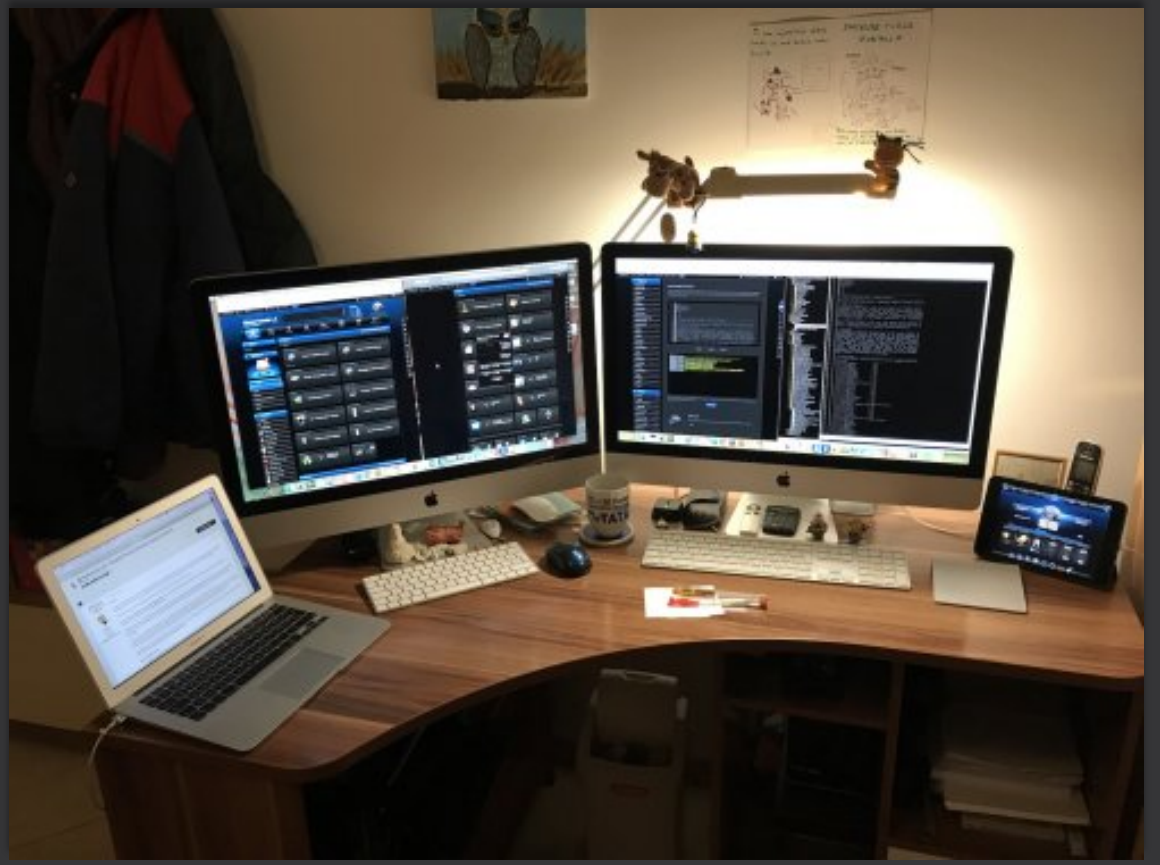
UHAS About Author

My name is Zoran Sankovic. I live in Croatia for almost last 50 years. I studied Chinese language and literature and graduated from Beijing language Institute in 1988. After that I studied electrical engineering at Faculty of maritime studies Rijeka. I worked 4 years for small IT company as program developer, service men and sales men and then decided that I need some change. I work now for almost 20 years for one offshore company. I'm happily married and have son and daughter, 12 and 14 years old.

I was always interested in home automation and I'm proud owner of Fibaro HC2 now for 4 years. It was really easy choice for me since Fibaro offers LUA programming language. Due to my work (5 weeks onboard/5 weeks at home) and my obligations to my family I had fairly little bit less than 1 year time to do all this programming. It would not be possible without help from Fibaro forum community!

UHAS roadmap is here. There are modules not explained in this presentation. Will do that in next one!

Sankotronic's Lab. Site



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